

used to display some headings or markings on the screen.

3. **Buttons:** These trigger events and are very essential components.
4. **Ball:** This is an inbuilt component in the App Inventor palette, which will display a ball on the screen. Using its properties, you can set the colour, diameter and various other parameters for the ball.
5. **Clock:** This component is responsible for all the time based events; so we will have a clock event also. The Clock component will repeat a particular action after a fixed duration of time.
6. **Horizontal arrangements:** These are special components which keep all child components horizontally aligned within them.

Table 1 gives the components that we will require for this application. We will drag them on to the designer from the palette on the left.

1. Drag and drop the components mentioned in the table to the viewer.
2. Visible components can be seen while the non-visible components will be located beneath the viewer under the tag 'Non-visible'.
3. We have placed a label on which to place the name of the application.
4. All buttons need to be put within the 'Horizontal arrangement' to keep them aligned horizontally.

5. If you have dragged and placed everything, the layout will look something like what's shown in Figure 1.
6. Make the necessary property changes, like we did when changing the text property for the label and button components.
7. Renaming the components helps to identify them in the block editor.
8. Your graphical user interface is now ready. Figure 1 shows exactly how the application will look after the installation.
9. Figure 2 gives the hierarchy of the components that we have dragged to the designer.

Now we will head towards the block editor to define the behaviour. Let's

|   | Component name         | Purpose                                      | Location                                  |
|---|------------------------|----------------------------------------------|-------------------------------------------|
| 1 | Canvas                 | To arrange the child components horizontally | Palette-->Drawing and Animation-->Ball    |
| 2 | Label                  | To display a label                           | Palette-->User Interface-->Label          |
| 3 | Ball                   | To receive input from the user               | Palette-->Drawing and Animation-->Ball    |
| 4 | Clock                  | To confirm the user's choice                 | Palette-->Sensors-->Clock                 |
| 5 | Button                 | To trigger events                            | Palette-->User Interface-->Button         |
| 6 | Horizontal arrangement | To arrange the child components              | Palette-->Layout-->Horizontal Arrangement |

Table 1

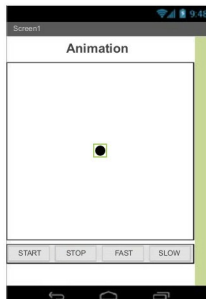


Figure 1: Designer screen



Figure 2: Components view

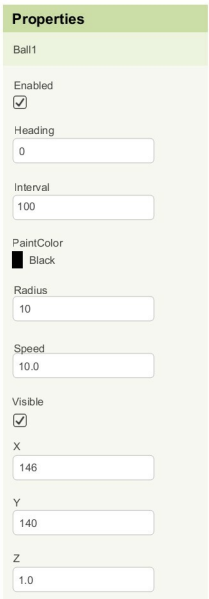


Figure 3: Block editor image 1